

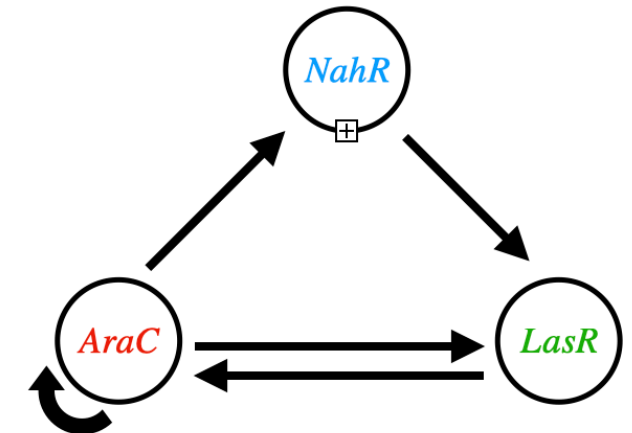
Biological Circuit Design

- Polycistronic Network Design for Insert:

$$P_{las} \rightarrow P_{bad} \rightarrow AraC \parallel P_{bad} \rightarrow NahR \parallel P_{bad} \rightarrow P_{sal} \rightarrow LasR$$

-----P1----- || -----P2----- || -----P3-----

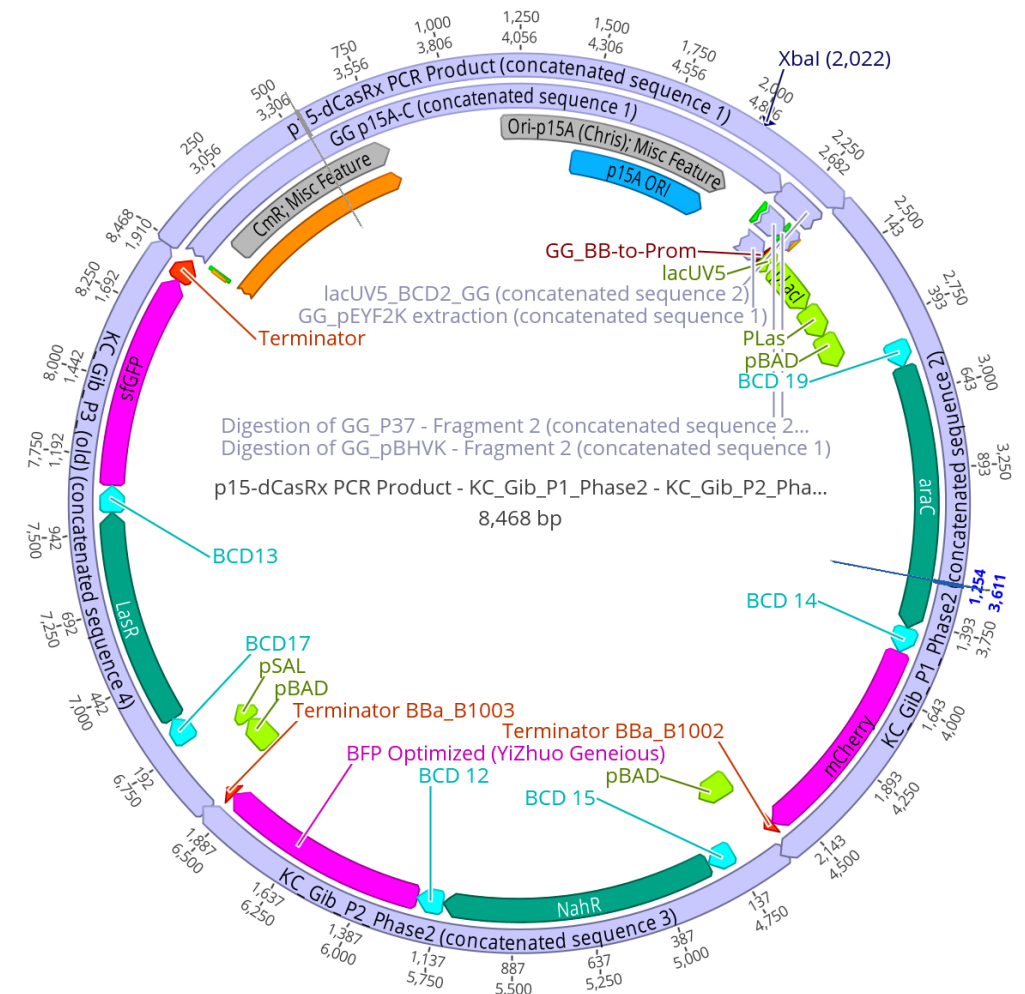
Hysteretic Robust Network Topology



Ribosome Binding Sites

- Middle Level strength to prevent overtaking cells

	LacI-36-GFP	AraC-36-GFP	RFP-36-GFP	RFP	Cell-36-GFP	Cell-36-RFP	TetR-36-GFP	TetR-full-GFP	PMK-36-GFP	PMK-36-RFP	PA-36-GFP	PA-36-RFP	GFP	GFP-36-RFP
BCD2	562.23	384.92	447.75	113.17	340.45	87.85	553.97	371.95	437.22	150.63	85.37	52.42	589.92	163.75
BCD1	494.72	275.21	462.31	105.26	314.55	69.11	409.98	321.45	435.43	122.38	86.08	37.81	462.95	144.18
BCD5	311.55	242.12	422.40	101.23	325.19	73.82	501.32	334.67	407.04	126.24	117.47	40.80	428.97	147.86
BCD7	259.49	269.01	400.00	99.82	324.57	75.24	515.33	363.32	415.72	126.14	115.46	43.32	451.36	148.21
BCD6	305.89	132.63	188.40	63.42	184.73	49.12	480.78	310.74	371.04	115.32	78.96	32.01	437.27	116.61
BCD11	271.45	196.13	331.76	80.88	159.92	48.00	489.17	365.69	256.80	64.41	60.75	28.57	323.44	94.37
BCD10	321.93	208.94	331.47	80.77	209.21	59.27	482.81	326.87	284.36	73.12	43.95	24.63	325.11	106.00
BCD9	489.81	314.98	381.18	88.15	341.57	58.18	551.78	351.78	267.88	88.18	88.71	88.71	368.68	115.11
BCD13	283.18	199.19	287.32	79.96	228.75	63.44	286.25	260.49	405.20	133.72	81.84	37.64	235.28	85.04
BCD15	359.83	159.32	293.25	65.28	153.44	47.53	362.12	381.42	534.14	114.51	55.35	28.77	168.84	60.61
BCD12	123.26	192.55	286.83	86.55	209.56	53.61	304.26	126.05	296.90	105.57	77.12	37.85	318.19	111.27
BCD17	92.53	155.52	191.26	64.56	152.93	42.95	258.86	176.90	300.29	116.03	66.77	35.94	207.58	75.02
BCD19	184.73	201.77	171.65	65.31	180.56	47.04	242.39	183.74	323.87	140.17	89.67	40.24	210.63	77.51
BCD14	80.47	179.96	238.37	63.92	217.82	58.45	362.49	326.43	297.08	101.25	67.25	32.77	188.74	82.72
BCD18	86.41	68.20	169.27	50.35	142.52	39.67	195.85	216.47	163.85	60.87	33.17	18.50	188.61	60.51
BCD20	63.78	184.28	151.49	49.26	94.42	31.04	164.78	185.91	130.11	57.33	24.56	17.19	133.18	42.32
BCD21	10.27	67.95	110.26	34.75	94.87	28.69	176.34	187.15	103.02	40.10	22.38	17.15	94.68	39.18
BCD16	92.66	60.15	33.70	27.45	43.78	12.42	64.38	119.56	77.93	39.03	9.48	9.22	61.16	13.34
BCD23	33.79	45.00	89.96	40.48	65.23	25.45	79.56	141.63	78.86	40.43	17.12	14.92	71.46	37.97
BCD24	11.91	33.22	39.77	19.02	35.20	14.80	46.21	74.11	41.88	11.06	12.44	11.53	53.99	20.55
BCD8	17.62	25.81	27.95	15.49	26.74	11.39	24.07	71.64	25.70	17.70	10.32	9.43	76.78	8.26
BCD22	15.87	10.28	8.62	7.28	11.68	7.63	23.22	36.79	19.48	11.67	6.04	6.87	28.27	6.78



Mutalik, Guimaraes, et al.

Biological Circuit Design

- Polycistronic Network Design for Insert:

$$P_{\text{las}} \rightarrow P_{\text{bad}} \rightarrow \text{AraC} \parallel P_{\text{bad}} \rightarrow \text{NahR} \parallel P_{\text{bad}} \rightarrow P_{\text{sal}} \rightarrow \text{LasR}$$

-----P1----- || -----P2----- || -----P3-----

Downstream Reporters

P1 - mCherry (source: snapgene)

P2 - tagBFP_ecoli_optimized (source: YiZhuo on Geneious)

P3 - sfGFP (source: snapgene)

Backbone

- Claire's p15d
 - Low copy number with chloramphenicol selection

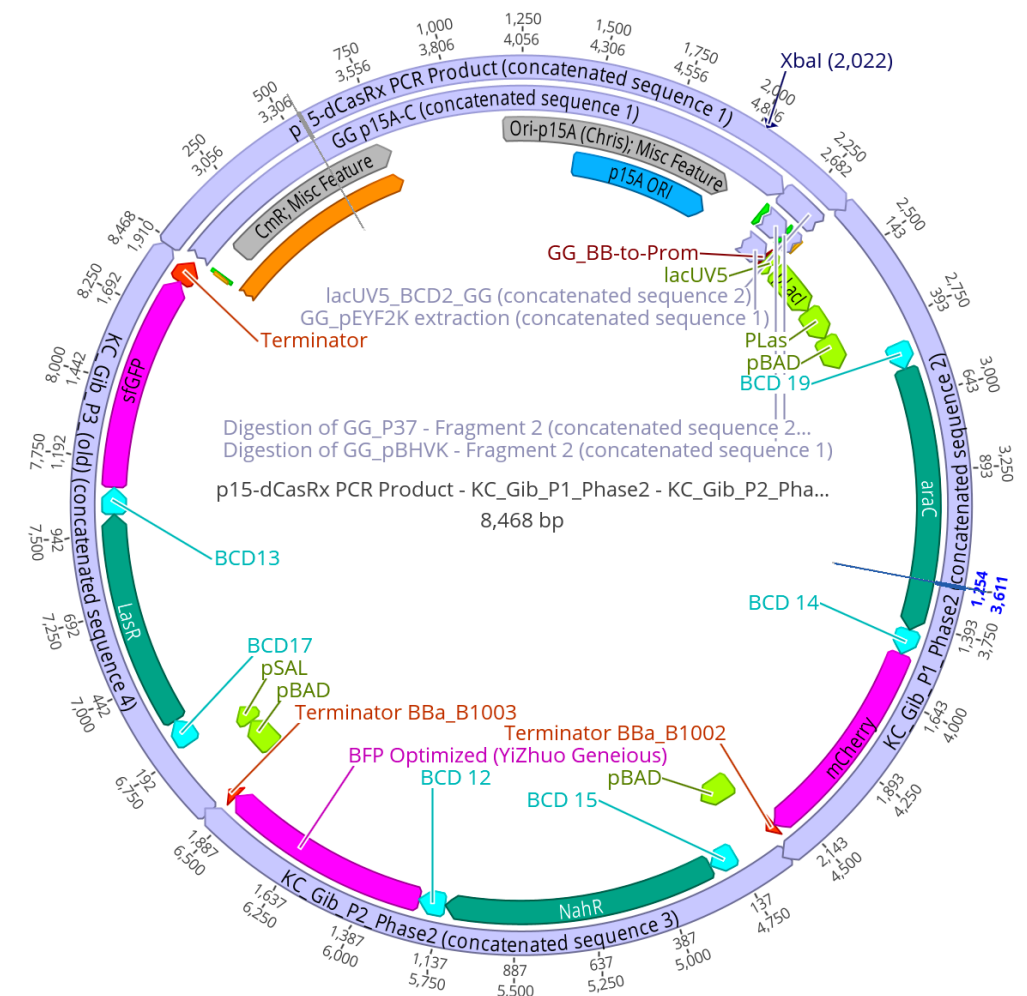
Terminators

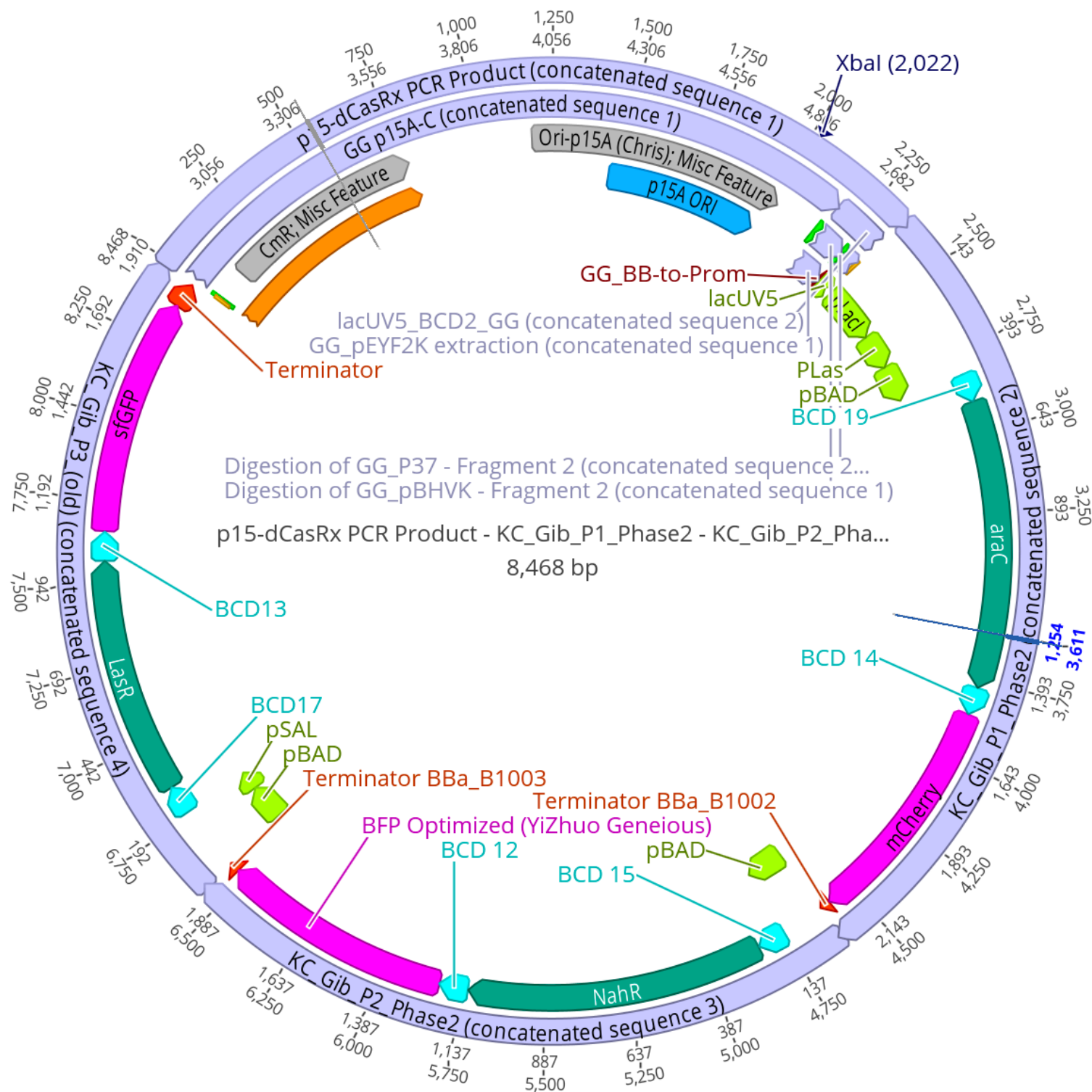
P1 - BBa B1002

P2 - BB1 B1003

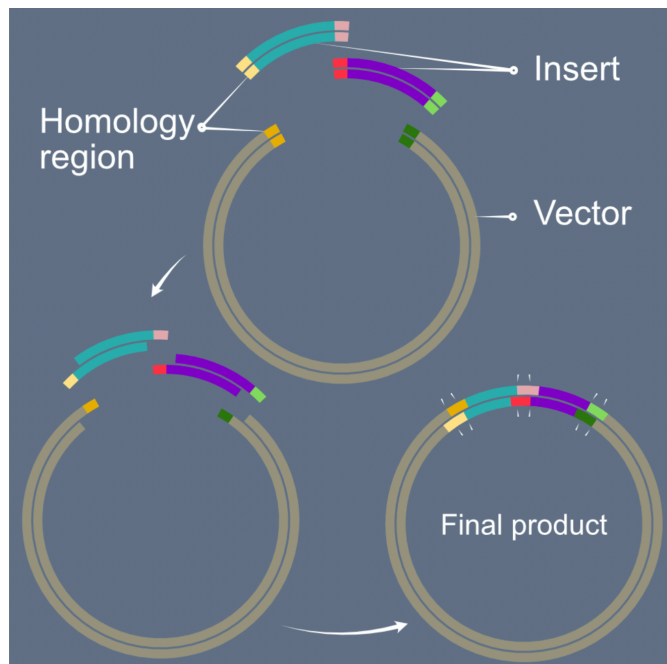
P3 - BBa BB0010

(sources: iGem)

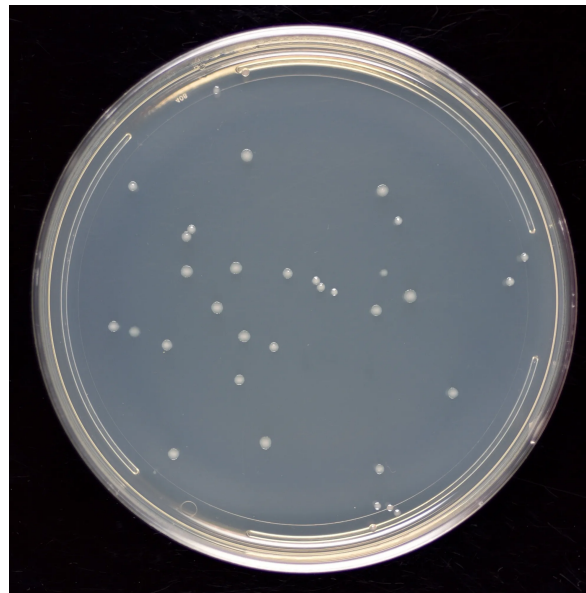
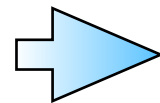




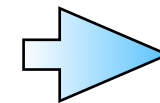
Project Workflow



Gibson Assembly



Transform and Pick Colonies



Validate Via Sequencing

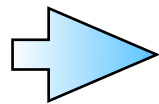
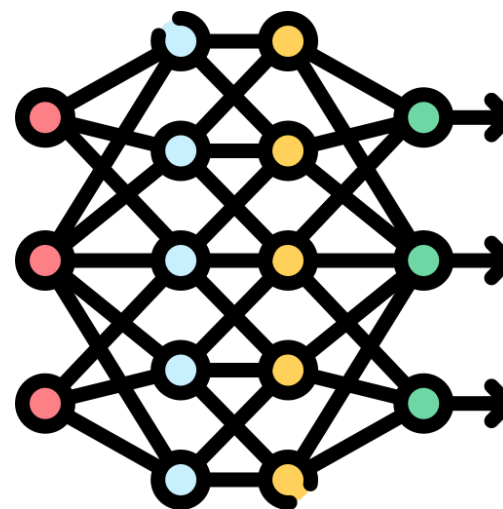
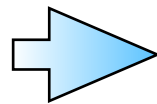


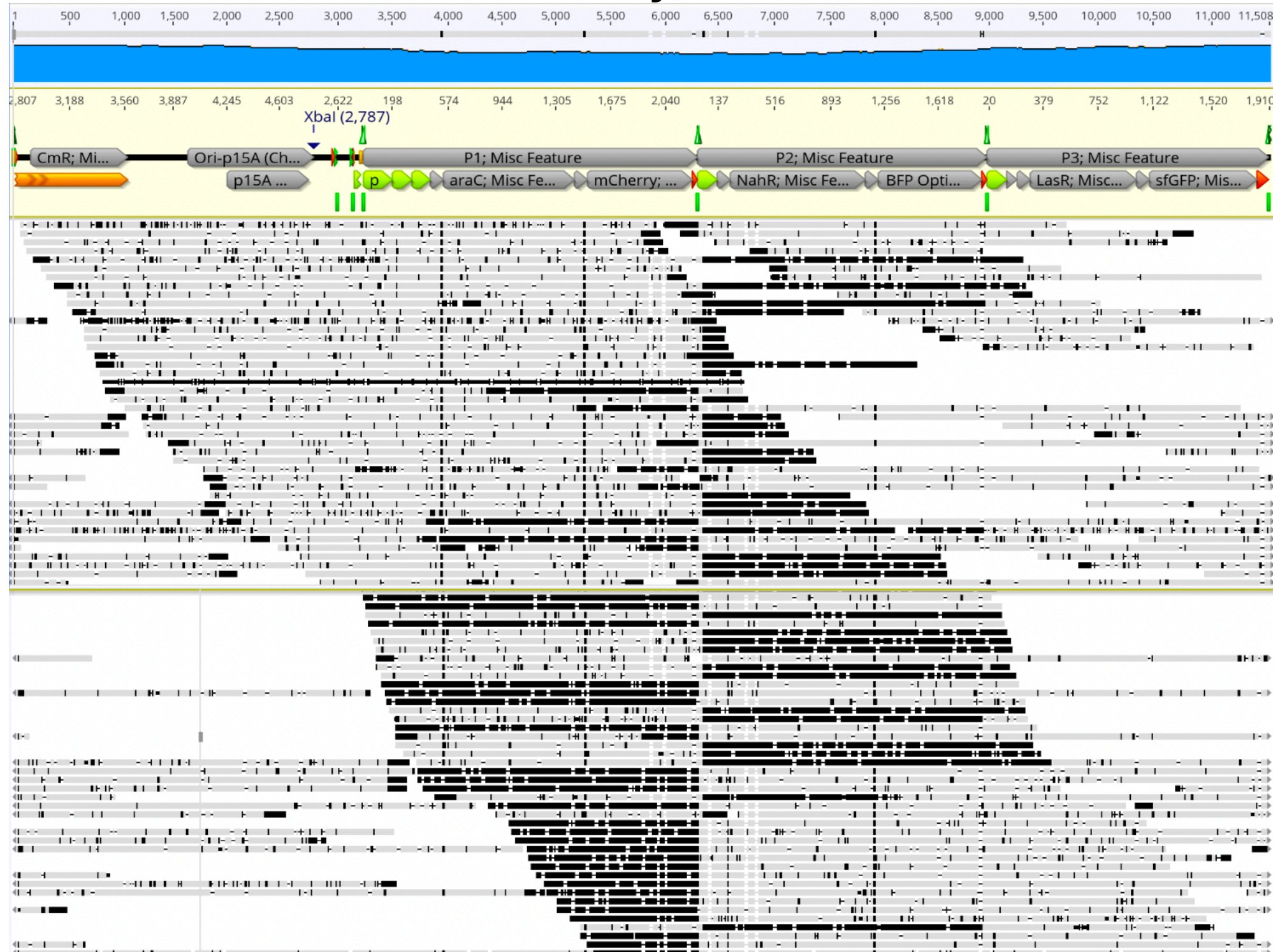
Plate Reader Experiments



Model Hysteretic Activity
& Param Fit using GANs

Successfully Validated Cloning

FASTAQ Sequencing Files are Mapped to Ref Sequence
Colony 5D:





Winter Group Meeting 2025



Update: Implementation of a Robust Hysteretic Gene Network in *E. coli*

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University of California, Santa Barbara

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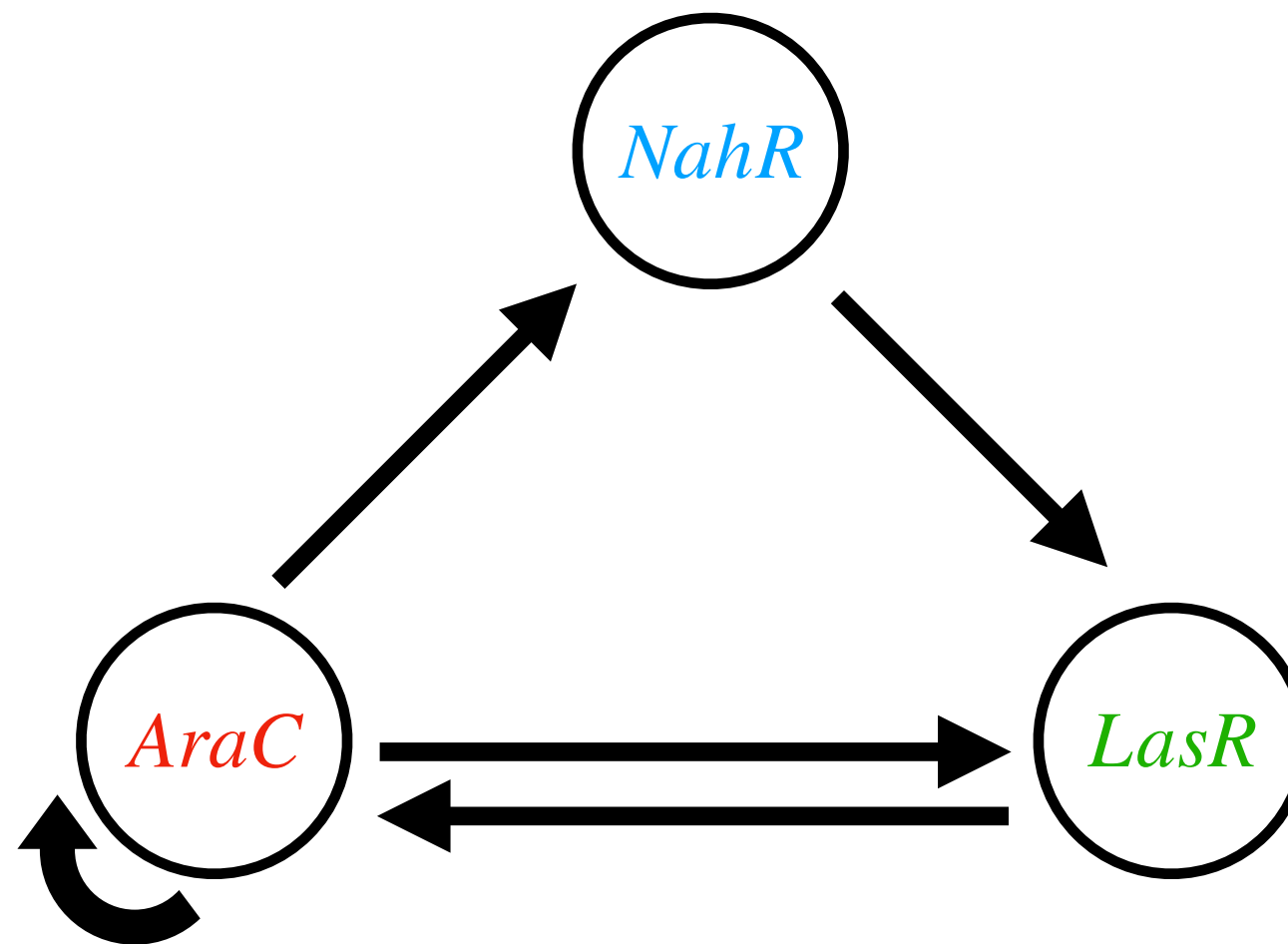


THE BIOLOGICAL CONTROL LAB

Theoretical Network Design

- Network Topology comes from Steve Haase and other collaborators
- Each arrow is an activator, each node is a gene
 - No inhibitors
- P1 is self activated

Hysteretic Robust Network Topology



Transformed into MG1655 Δ LacI (Experimental Comp Cells)

Colony PCR:

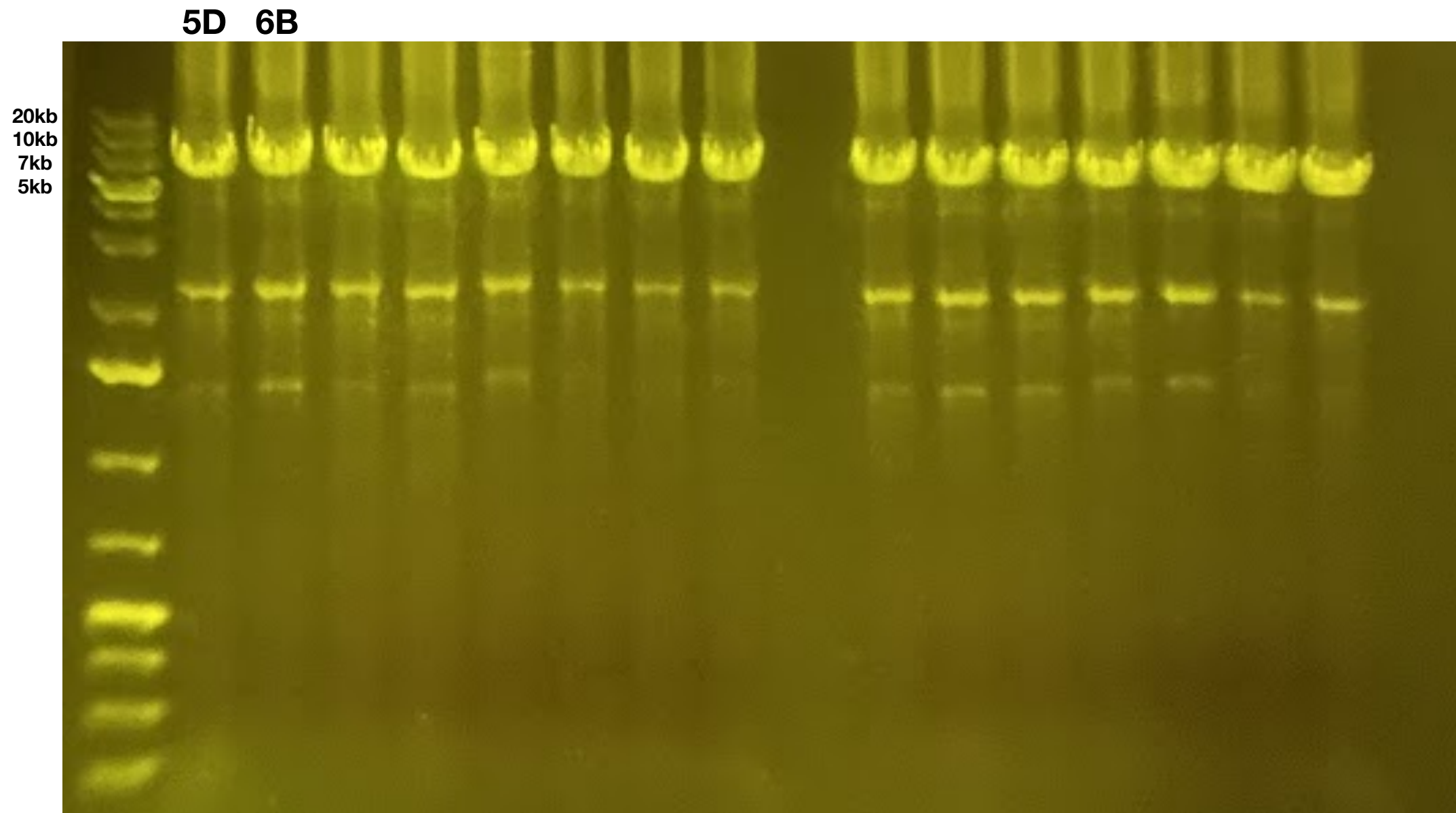
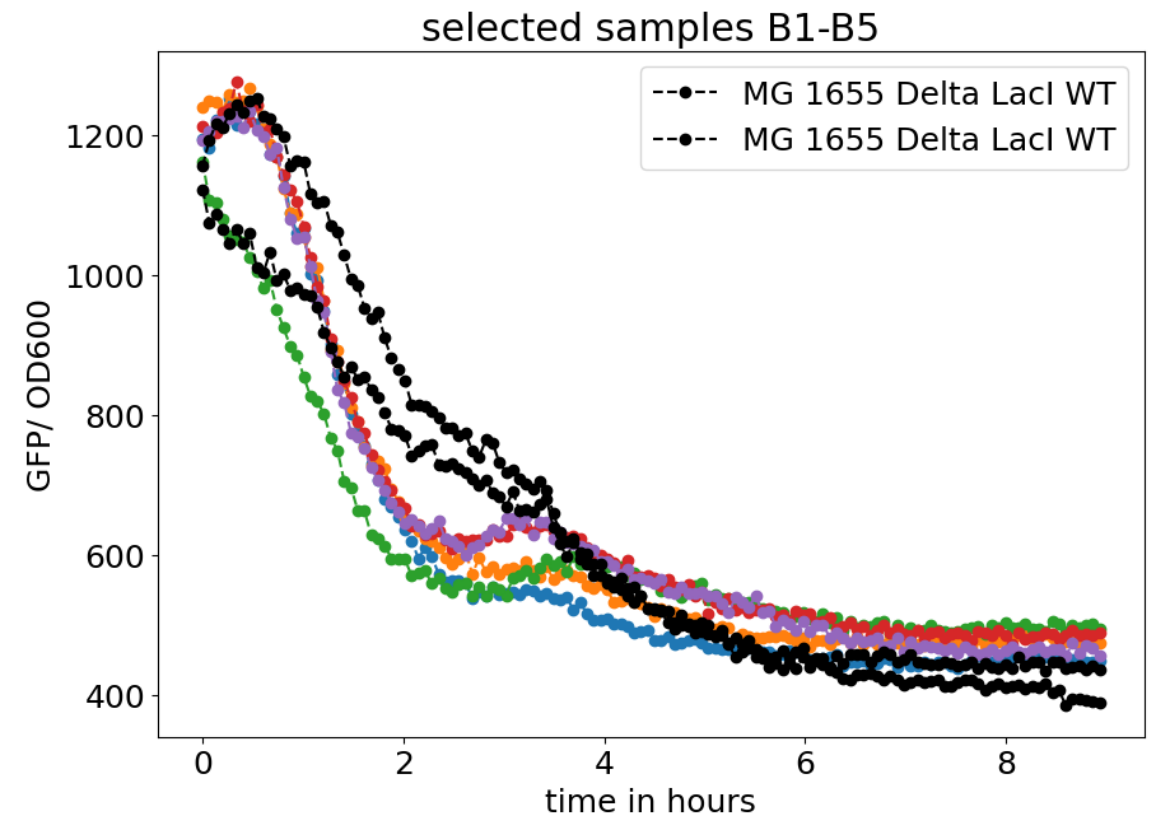
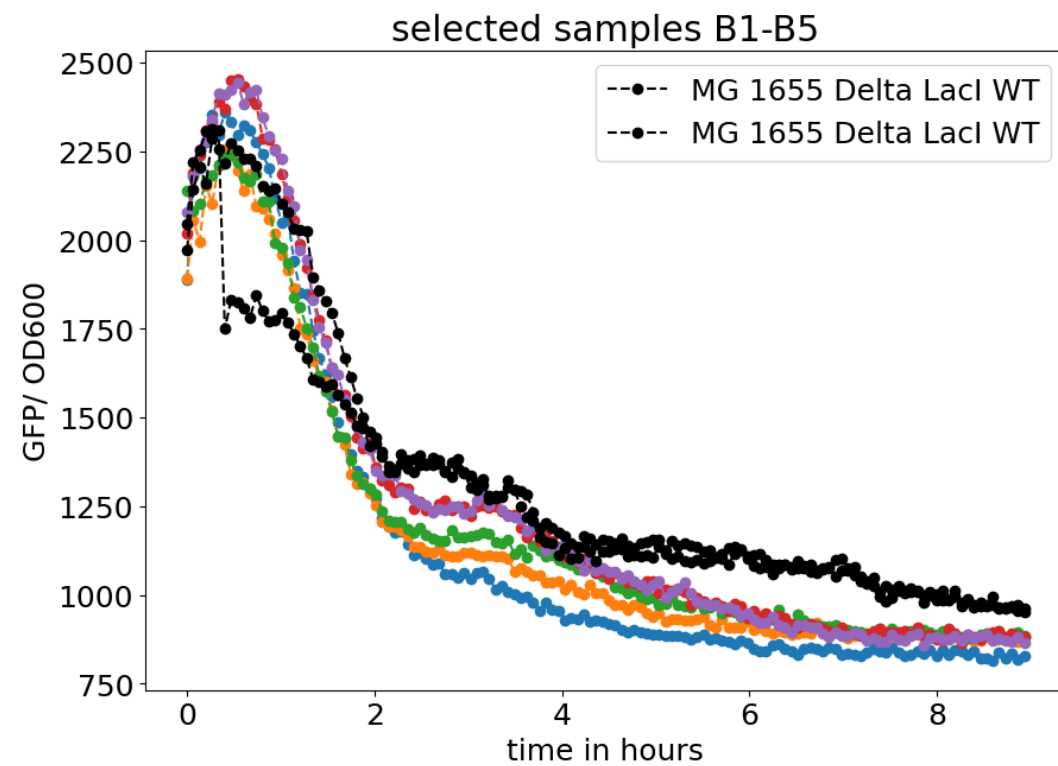
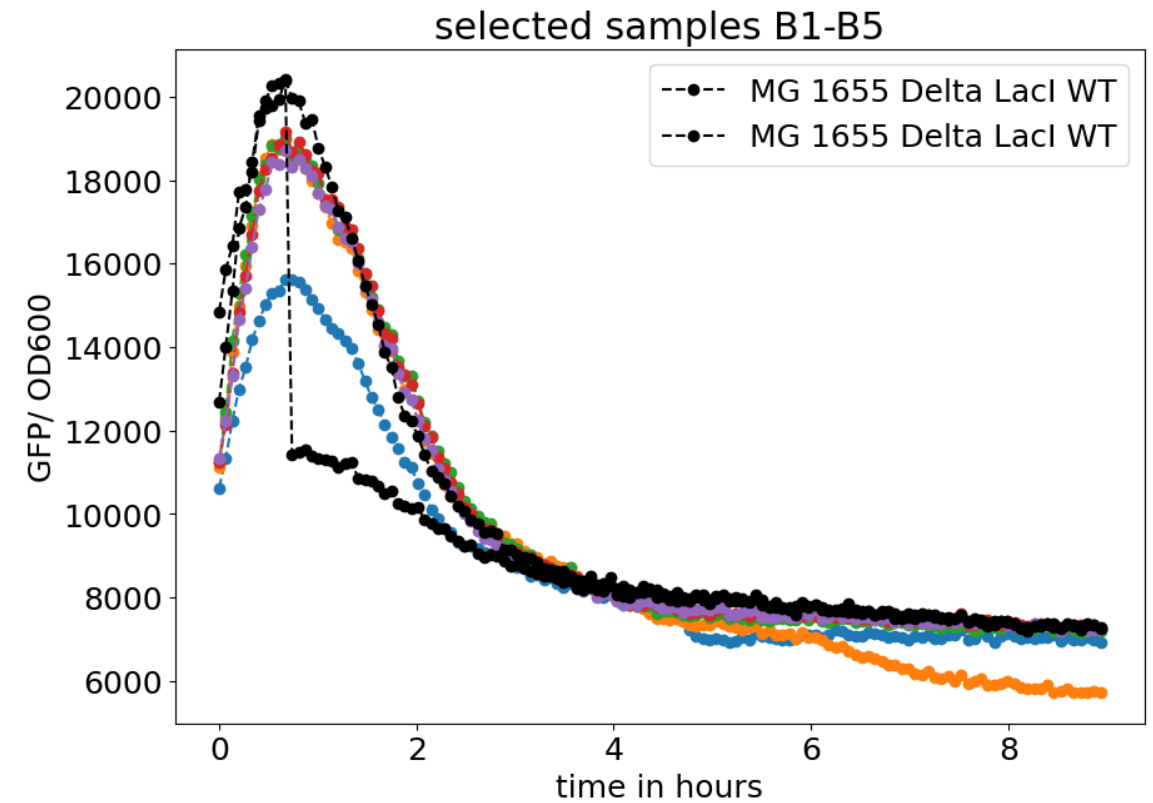
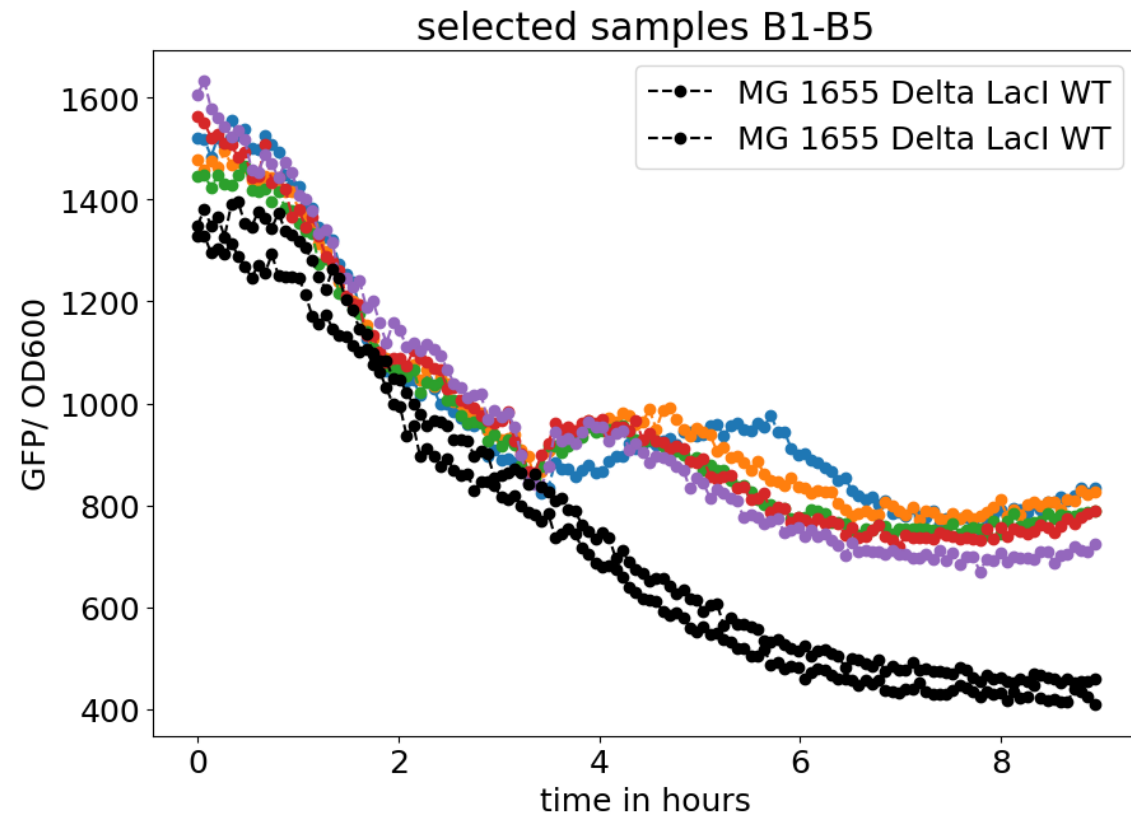


Plate Reader Setup

96 Well Plate												
	1	2	3	4	5	6	7	8	9	10	11	12
A												
B	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Exp (No Inducer)
C	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Arab	Exp
D	HSL	HSL	HSL	HSL	HSL	HSL	HSL	HSL	HSL	HSL	HSL	Exp
E	HSL	HSL	HSL	HSL	HSL	HSL	HSL	HSL	HSL	HSL	HSL	Exp
F	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Exp
G	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Sal	Exp
H											WT	MG1655 WT
Serial Dilution L to R -->												
Highest Conc. Inducer											Lowest Conc. Inducer	
Experimental Wells are in Chlor-34 LB												
Wildtype Wells are in No Selection LB												
Row 1-2: I add 0.75uL of 20% Arabinose to 250uL LB to reach 4mM.												
Row 3-4: I add 2.5uL of 10mM of Sal in Water to 250uL LB to reach 100uM												
Row 5-6: I add 2.5uL of 1mM HSL in Eth to 250uL LB to reach 10uM												

Plate Reader Data

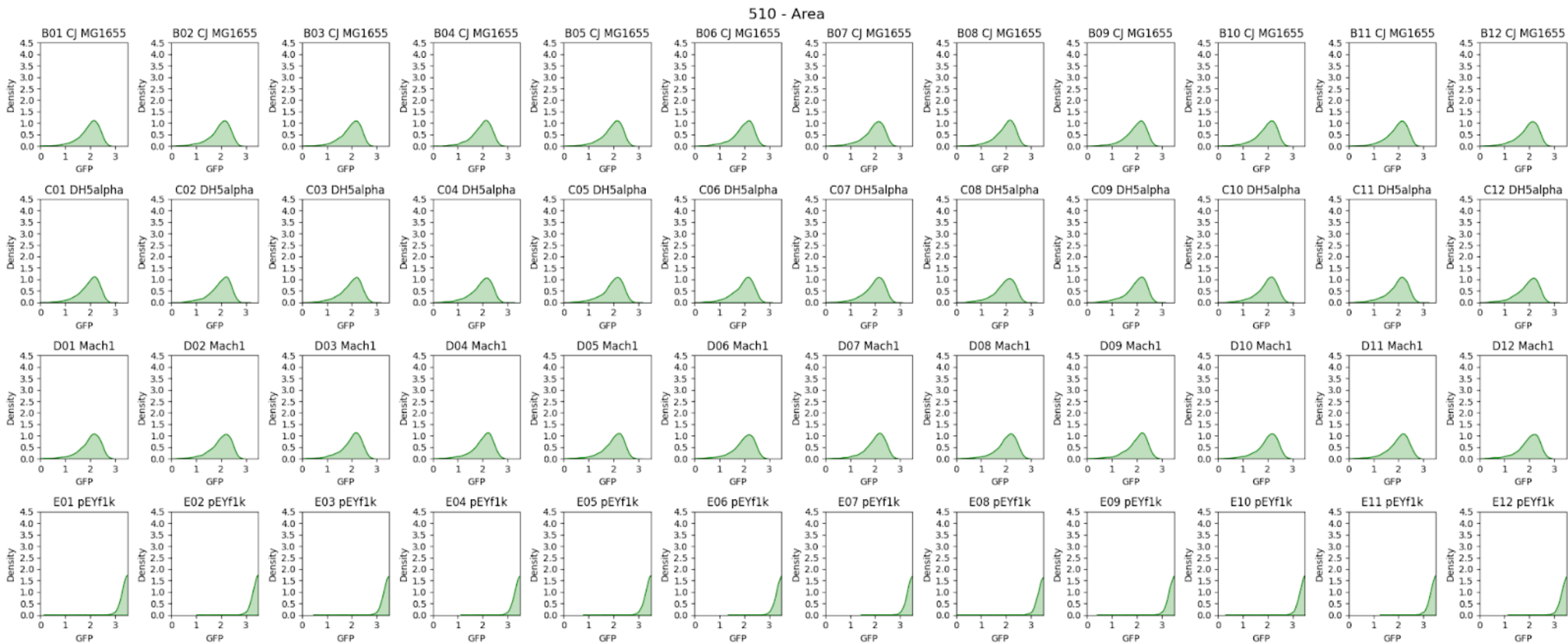


Flow Cytometry Data



Testing WT Data

The activity in empty WT strains is lower in GFP relative to the positive control



Next Steps

Rerun Data

Take out Plates at right time

Modeling with Hill Functions

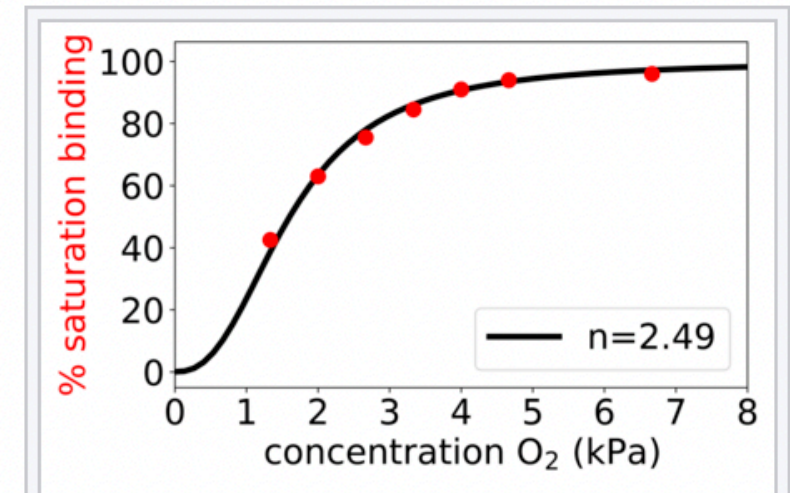
The Hill equation is commonly expressed in the following ways.^{[2][7][8]}

$$\begin{aligned}\theta &= \frac{[L]^n}{K_d + [L]^n} \\ &= \frac{[L]^n}{(K_A)^n + [L]^n}, \\ &= \frac{1}{1 + \left(\frac{K_A}{[L]}\right)^n}\end{aligned}$$

where:

- θ is the fraction of the **receptor protein** concentration that is bound by the **ligand**,
- $[L]$ is the total **ligand** concentration,
- K_d is the apparent **dissociation constant** derived from the **law of mass action**,
- K_A is the ligand concentration producing half occupation,
- n is the Hill coefficient.

[Wikipedia](#)



Plot of the % saturation of oxygen binding to haemoglobin, as a function of the amount of oxygen present (expressed as an oxygen pressure). Data (red circles) and Hill equation fit (black curve) from original 1910 paper of Hill.^[6]

Derived Mathematical Model

U_1 : inducer activates p_1 and is represented by u

a_k, d_k, n_k : activation, degradation, Hill coefficients

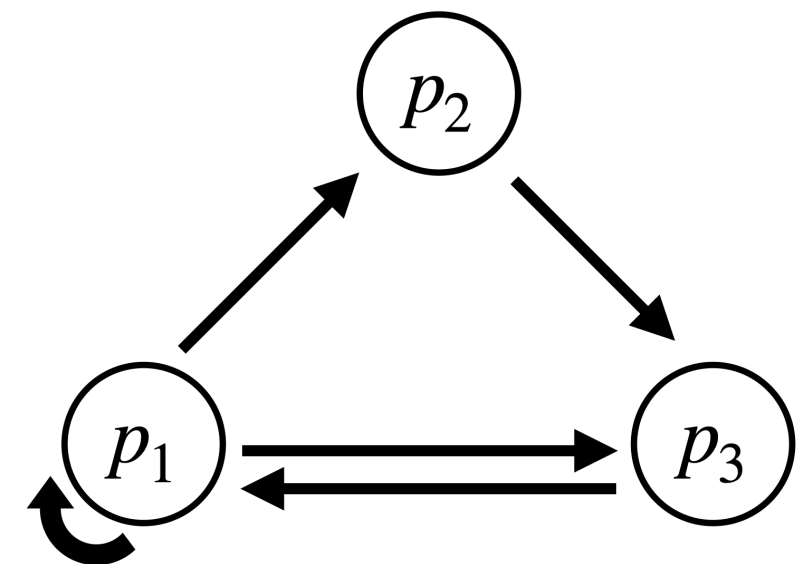
$$U_1 = \frac{u^{n_4}}{1 + u^{n_4}}$$

$$\frac{dp_1}{dt} = U_1 + a_1 \left[\frac{p_1^{n_1}}{1 + p_1^{n_1}} + \frac{p_3^{n_3}}{1 + p_3^{n_3}} \right] - d_1 p_1$$

$$\frac{dp_2}{dt} = a_2 \left[\frac{p_1^{n_1}}{1 + p_1^{n_1}} \right] - d_2 p_2$$

$$\frac{dp_3}{dt} = a_3 \left[\frac{p_1^{n_1}}{1 + p_1^{n_1}} + \frac{p_2^{n_2}}{1 + p_2^{n_2}} \right] - d_3 p_3$$

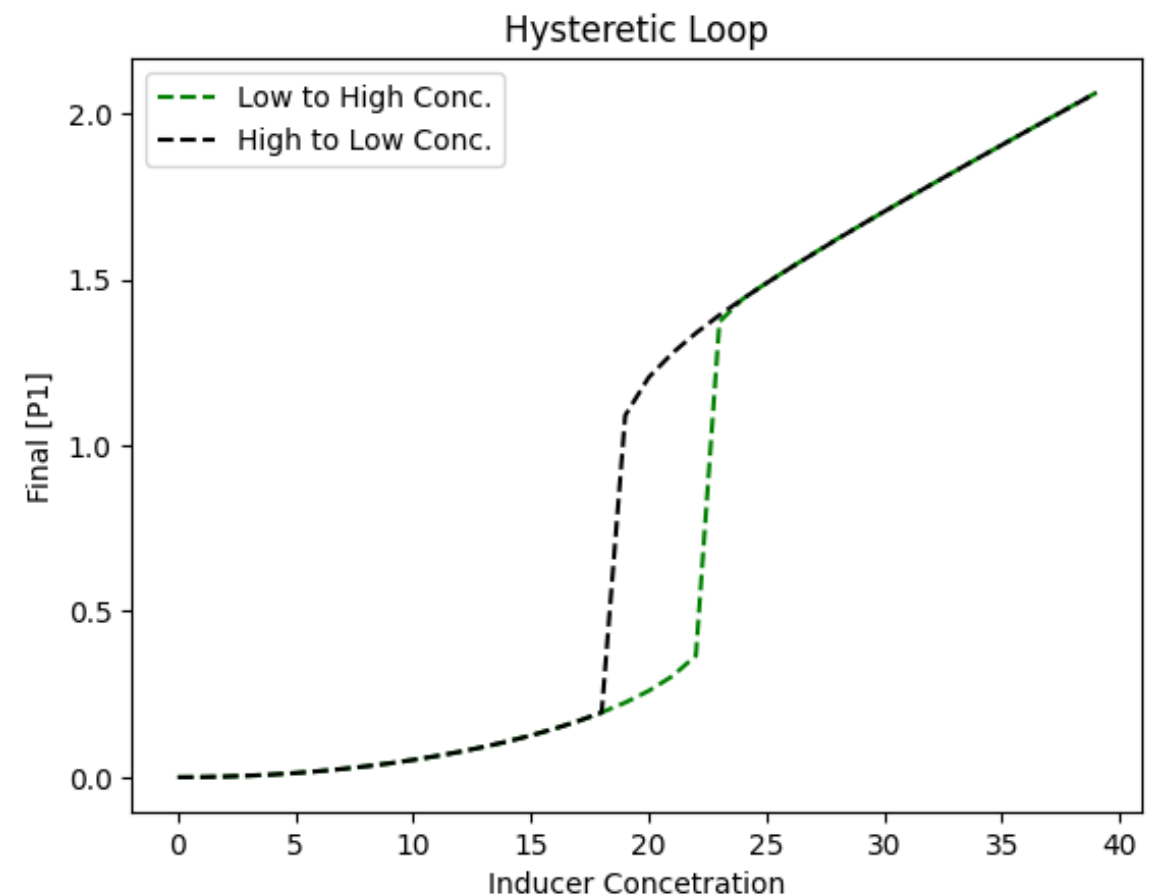
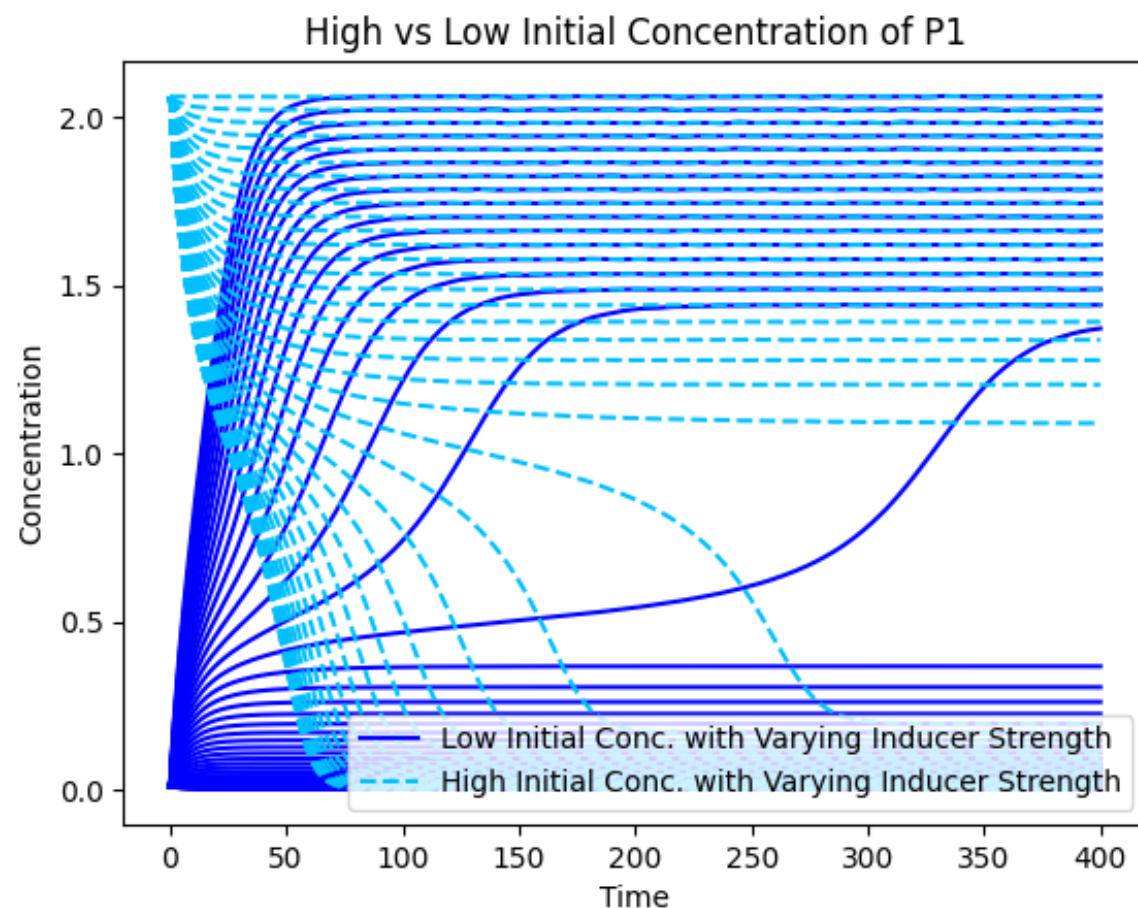
Hysteretic Robust Network Topology



Check for Hysteresis

- Each line carries a different inducer value u
- **Dark Blue:** Initial concentrations of p_1, p_2, p_3 is 0.01,0.01,0.01
- **Light Blue:** Initial concentrations of p_1, p_2, p_3 is highest final concentration
- Different final concentrations show bistability
- Parameters were chosen to show hysteresis (not all parameters show hysteresis)

```
params = [0.2, 0.2, 0.2, 0.2, 0.2, 0.2, 2,2,2] # Example values for a1, a2, a3, d1, d2, d3, n1, n2, n3  
inducers = ['X', 2] # Initialize inducers with values for u, n4. u has placeholder value 'X'
```





Thank you
Discussion / Q & A

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